HEALTHCARE DATA ANALYTICS

WEEK 01 PROJECT REPORT

Exploratory Analysis of Diabetes Indicators in Public Health Survey Data

Report detail	Information
Internship	Healthcare Data Analyst Intern — YuvaIntern
Intern	Rehan Raza Shaikh
Week	Week 01 — Project Planning, Data Familiarisation and Initial Analysis
Tools used	Python, pandas, Matplotlib, Power BI Desktop
Dataset	Diabetes Health Indicators Dataset derived from CDC BRFSS 2015
Report status	Initial implementation report; analysis and dashboard are in progress

1. Executive Summary

This project explores relationships between diabetes status and selected health indicators in a public survey-derived dataset. The initial work focused on organising the project, loading the CSV file in Python, inspecting the dataset, generating exploratory charts, and importing the data into Power BI to begin a dashboard.

The current work is an exploratory analysis, not a diagnostic or predictive clinical system. The survey data can describe patterns and associations, but it cannot establish causation or determine an individual's medical status.

Work completed so far

- Downloaded and loaded the diabetes health indicators CSV dataset.
- Created a Python script (analysis.py) using pandas and Matplotlib.
- Inspected the dataset structure, column names, missing values and exact duplicate rows.
- Generated charts for diabetes category distribution, average BMI by diabetes status, and diabetes-status distribution by high blood pressure.
- Imported the CSV into Power BI Desktop and created initial visuals for diabetes status, BMI, blood pressure, cholesterol and age.

Current status and next improvements

The Power BI page is an initial draft. In the submitted screenshot, Diabetes_012 appears to be treated as a continuous numeric field in some visuals (for example, category labels such as 0, 10 and 20 appear in the legend). Before presenting the dashboard as final, the field should be treated as a categorical field and the visual axes, aggregation and titles should be corrected. The age chart also needs checking because Age is an encoded age-group variable, not an exact age.

2. Project Overview

Project title: Exploratory Analysis of Diabetes Indicators in Public Health Survey Data.

Problem statement: Diabetes is an important public-health concern. Public survey data can be explored to understand how reported diabetes categories vary alongside selected demographic and health indicators. The project aims to summarise these patterns in clear charts and an interactive Power BI report.

Project scope

- Describe the distribution of survey records across the three Diabetes_012 categories.
- Compare average BMI across the categories.
- Explore reported high blood pressure and high cholesterol alongside diabetes status.
- Explore category counts across encoded age groups.
- Develop a clear Power BI dashboard with appropriate labels and filters.

Out of scope for this stage: clinical diagnosis, causal claims, treatment recommendations, or a validated machine-learning prediction model.

3. Dataset and Data Sources

The selected file is diabetes_012_health_indicators_BRFSS2015.csv, a ready-to-use dataset derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS). It contains survey records and health-indicator variables including Diabetes_012, BMI, HighBP, HighChol and Age.

Field	Meaning / interpretation used in this project

Diabetes_012	Target category: 0 = no diabetes, 1 = prediabetes, 2 = diabetes.
BMI	Body Mass Index value recorded in the survey dataset.
HighBP	Binary indicator for reported high blood pressure (0/1).
HighChol	Binary indicator for reported high cholesterol (0/1).
Age	Encoded age-group category (1–13), not exact age in years.

Primary dataset access: Kaggle, Diabetes Health Indicators Dataset (Alex Teboul):
https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset

Underlying source and documentation: CDC BRFSS Annual Survey Data — 2015:
https://www.cdc.gov/brfss/annual_data/annual_2015.html

Note: The report describes the downloaded file and its current exploratory use. Dataset documentation should be consulted before making detailed claims about survey design, weighting, representativeness or variable definitions.

4. Initial Data Inspection

Inspection item	Initial result / action
Rows	253,680 records in the downloaded dataset.
Columns	22 columns in the CSV file.
Missing values	The initial script reported zero missing cells in the loaded columns.
Exact duplicate rows	The initial script reported 23,899 exact duplicate rows.
Data types	Column data types were printed in Python for review.

Duplicate rows have not been automatically removed. Repeated records may reflect data characteristics or preprocessing decisions, so they must be investigated and documented before any removal. The original CSV should be preserved as a raw input.

5. Objectives and Deliverables

Objective	Deliverable / evidence
Understand the dataset	Python inspection output: shape, sample rows, column names, data types and missing-value check.
Summarise diabetes categories	Bar chart of survey-record counts by Diabetes_012 category.
Explore BMI patterns	Bar chart of average BMI by diabetes category.
Explore high blood pressure	Grouped percentage comparison of diabetes status for HighBP = 0 and HighBP = 1.
Start interactive visual reporting	Initial Power BI report page with category, BMI, blood pressure, cholesterol and age visuals.
Document responsible use	Limitations, privacy and interpretation notes in the report.

6. Initial Findings from the Generated Visuals

The following observations are preliminary and based on the screenshots generated during the initial analysis. Values should be confirmed from the Python output before being quoted as final results.

- The diabetes-status distribution is imbalanced: the no-diabetes category is the largest, the diabetes category is smaller, and the prediabetes category is the smallest.
- The average-BMI chart shows higher average BMI in the diabetes category than in the no-diabetes category, with prediabetes between them in the current chart.

- The high-blood-pressure comparison shows a larger proportion of records in the diabetes category among respondents coded as having high blood pressure than among those coded as not having high blood pressure.

These patterns are descriptive associations in survey data. They should not be interpreted as proof of causation or as clinical advice.

7. Evidence: Python Analysis

The following screenshots document the charts generated in Python. They are included as evidence of work completed during the initial analysis stage.

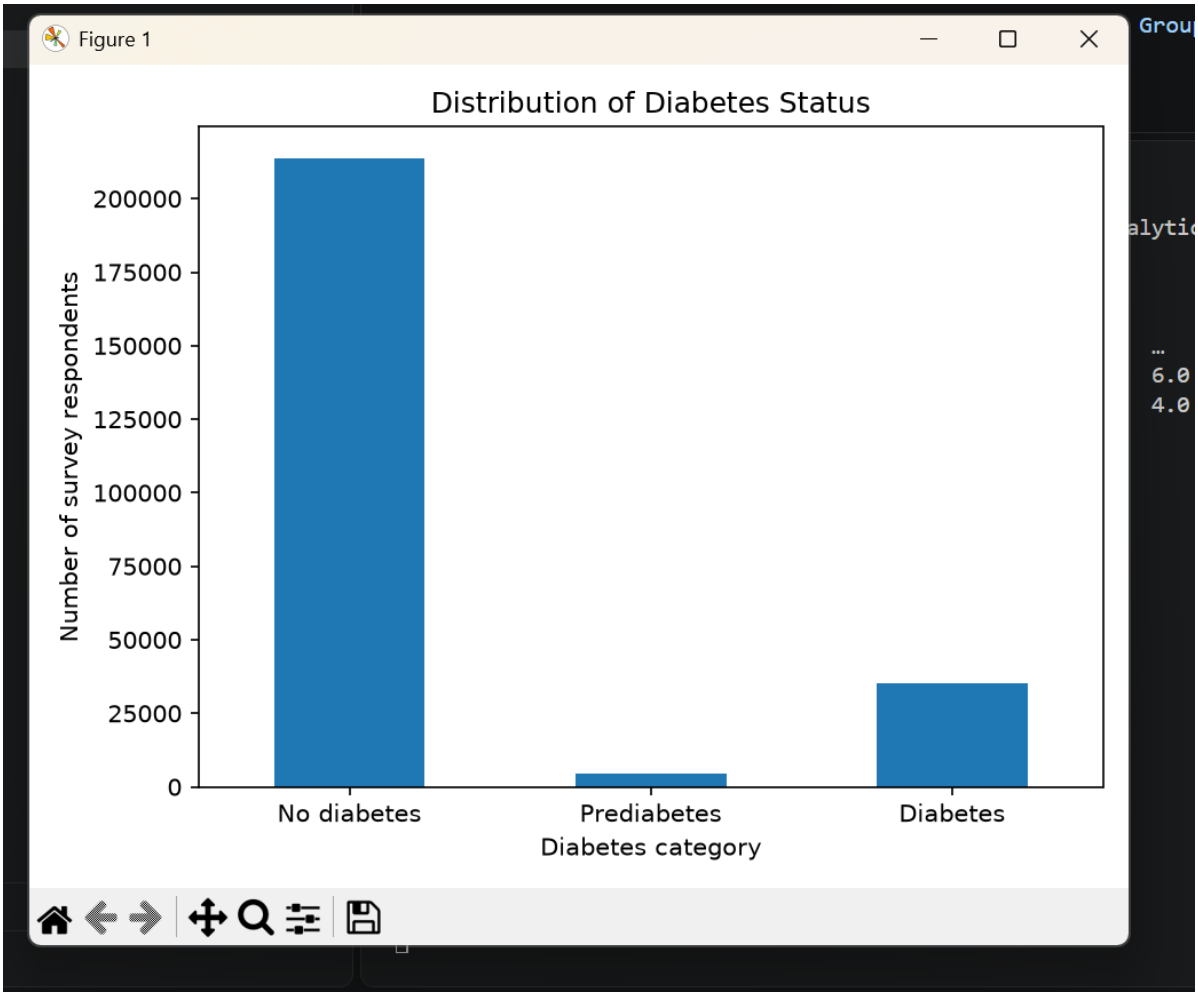


Figure 1. Distribution of diabetes status generated using Python and Matplotlib.

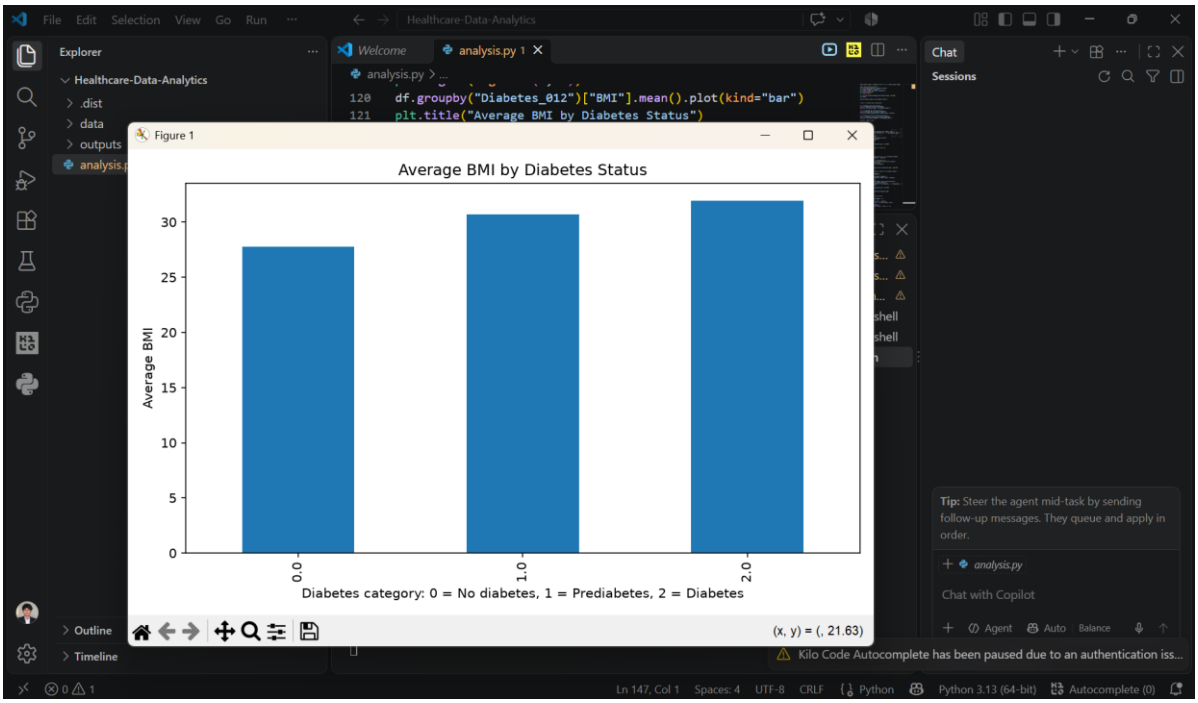


Figure 2. Average BMI by diabetes status generated using Python and Matplotlib.

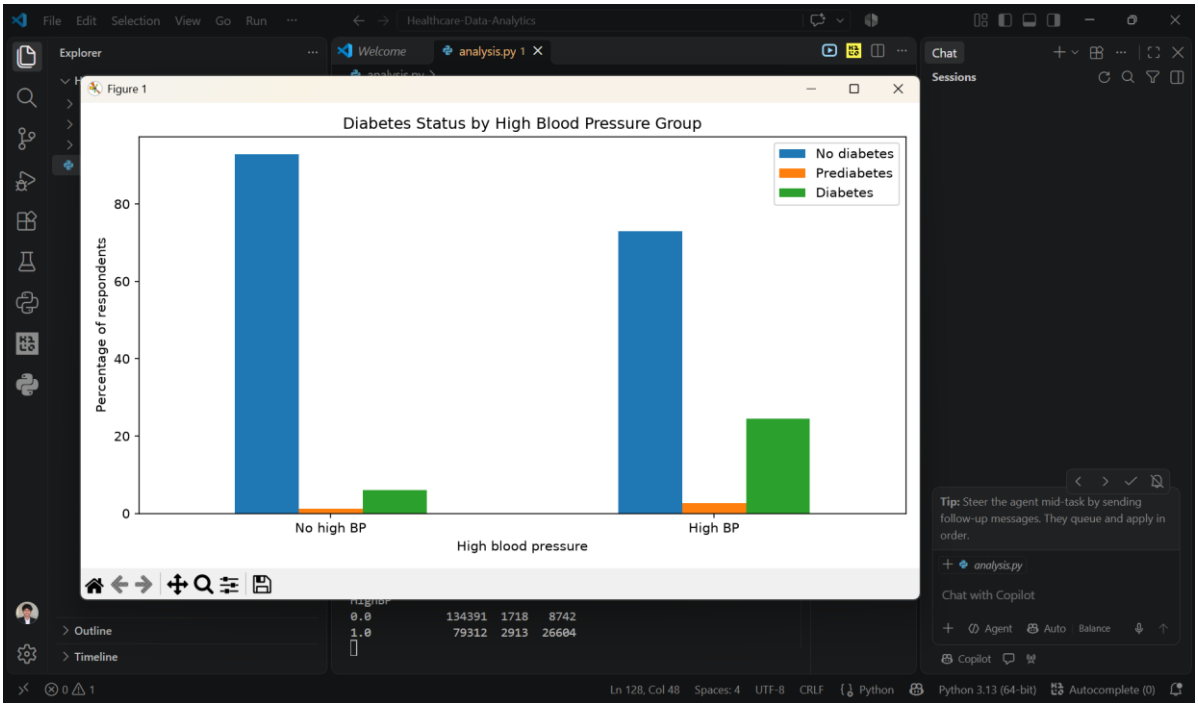


Figure 3. Diabetes-status distribution by high-blood-pressure group (percentages within each group).

8. Evidence: Power BI Initial Dashboard

The screenshot below shows the first draft of the Power BI report page after importing the CSV. It includes initial category distribution, blood-pressure comparison, age-group comparison, BMI, cholesterol and donut visuals. Visual field settings and category labels still require refinement.

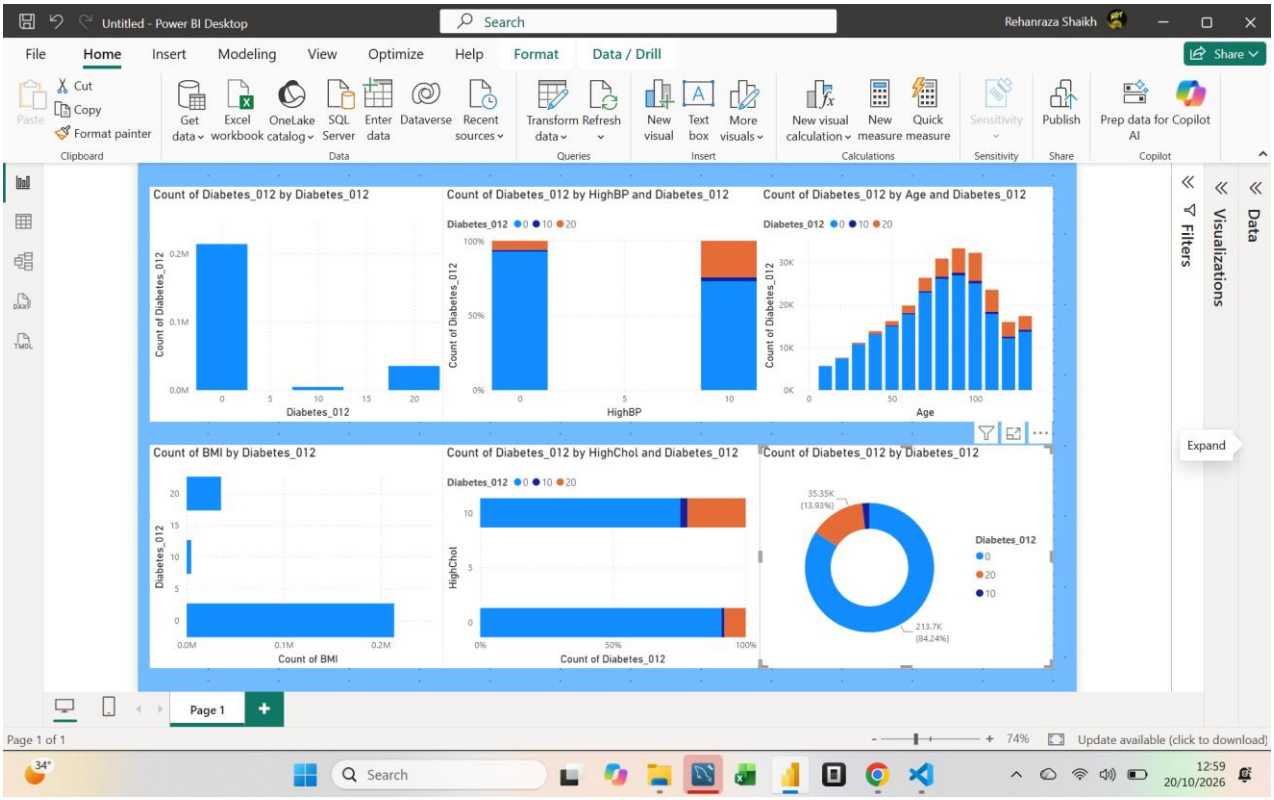


Figure 4. Initial Power BI dashboard draft. This is a work-in-progress report page.

9. Planned Power BI Visuals and Field Mapping

Visual	Field placement	Purpose / note
Card: Total records	Values: Diabetes_012 → Count	Counts survey records, not necessarily unique individuals.
Clustered column: category distribution	X-axis: Diabetes_012 (categorical); Y-axis: count of Diabetes_012	Show three discrete categories 0, 1 and 2.
Bar/column: average BMI	Category axis: Diabetes_012; Values: BMI → Average	Compare average BMI by diabetes category.
100% stacked column: HighBP comparison	X-axis: HighBP; Values: count of Diabetes_012; Legend: Diabetes_012	Compare within-group proportions; category field must be categorical.
100% stacked bar: HighChol comparison	Category axis: HighChol; Values: count of Diabetes_012; Legend: Diabetes_012	Compare within-group proportions.
Stacked column: age group	X-axis: Age; Values: count of Diabetes_012; Legend: Diabetes_012	Use encoded age-group order and a clear label.
Slicers	Age, HighBP, HighChol	Enable filtering across visuals.

Power BI correction note: Set Diabetes_012 to a categorical/discrete field for category-based charts. Check each visual's aggregation and axis type. Avoid using latitude/longitude or a map because the selected dataset does not provide geographic coordinates for this analysis.

10. Project Timeline

Week	Planned activity	Expected output
Week 1	Project planning, dataset acquisition, initial Python inspection and first charts; begin Power BI import.	Project plan, initial code, first charts and draft dashboard.
Week 2	Data dictionary review, investigate duplicates, verify coding and	Data-quality notes and reproducible

	document cleaning decisions.	cleaning steps.
Week 3	Exploratory analysis across BMI, blood pressure, cholesterol, age group and other available indicators.	Additional charts and documented observations.
Week 4	Refine Power BI layout, labels, slicers, measures and interactions.	Usable interactive dashboard draft.
Week 5	Validate results against Python summaries, improve report and document limitations.	Reviewed dashboard and analysis summary.
Weeks 6–8	Optional extension if approved: deeper analysis, stakeholder feedback, final documentation and presentation.	Final project package and presentation.

The timeline is a proposed plan and may be adjusted to match the internship's official submission schedule.

11. Risk Assessment and Contingency Planning

Risk	Impact	Mitigation / contingency
Incorrect field aggregation in Power BI	Charts may show misleading numeric categories or totals.	Set category fields to categorical; verify Count/Average settings; cross-check with Python.
Duplicate records	May affect counts and summary statistics.	Investigate exact duplicates, retain raw source, document rationale before cleaning.
Misinterpretation of survey associations	Could lead to unsupported causal or clinical claims.	Use descriptive wording and clearly state limitations.
Large CSV or slow Power BI performance	Long load times or slow visuals.	Close unused applications; start with a small number of visuals; use Power Query only as needed.
Unclear variable coding	Labels or comparisons could be incorrect.	Review the dataset's data dictionary and source documentation.
Accidental loss of raw data	Analysis becomes hard to reproduce.	Keep an unchanged copy of the downloaded CSV and save cleaned outputs separately.

12. Ethical Considerations and Limitations

- Use public, de-identified survey-derived data only for the stated learning and analytics purpose.
- Do not attempt to identify respondents or combine the data with identifying information.
- Avoid diagnostic claims or personal medical recommendations based on the charts.
- Present group differences as associations, not causes.
- Check dataset documentation before claiming that results are nationally representative or survey-weighted.
- Acknowledge that the target categories are imbalanced and that raw counts alone can be misleading; percentage comparisons may be more appropriate for some charts.
- Keep the report reproducible by retaining the raw dataset, source links, Python script and Power BI file.

13. Resources Used

Resource	Use
Python 3	Running the analysis script.
pandas	Loading, inspecting and summarising tabular data.
Matplotlib	Generating exploratory bar charts.
VS Code	Editing and running analysis.py.
Power BI Desktop	Importing data and creating an interactive dashboard draft.
Kaggle dataset and CDC documentation	Dataset access and source/variable reference.

No monetary resource estimate is included because no software, hardware or service costs were provided. The listed tools were used as project resources.

14. Quality Assurance Checklist

- Confirm CSV row and column counts against the loaded file.
- Verify Diabetes_012 codes and use clear category labels.
- Check whether BMI uses Average rather than Sum in Power BI.
- Ensure percentage charts use within-group percentages where intended.
- Investigate duplicates without automatically deleting them.
- Use clear chart titles, axis titles, readable labels and consistent formatting.
- Cross-check Power BI values against Python outputs.
- Save the PBIX file and preserve the raw CSV and Python script.

15. Conclusion and Next Steps

Week 01 work established the basic analysis workflow: data acquisition, Python inspection, exploratory chart generation and the start of a Power BI report. The next priority is to correct categorical field handling in Power BI, review the dataset's variable definitions, investigate duplicates, and validate key summaries against Python.

After these checks, the dashboard can be improved with summary cards, clearly labelled charts and interactive slicers. The final report should present only verified findings and should retain the limitations of observational survey data.

Appendix A — Files produced during the work

- analysis.py — Python script used for dataset inspection and exploratory charts.
- outputs/diabetes_distribution.png — initial diabetes-status distribution chart.
- outputs/average_bmi.png — average BMI chart.
- outputs/blood_pressure_comparison.png — blood-pressure comparison chart.
- Diabetes_Healthcare_Analytics.pbix — planned Power BI report filename; confirm the file is saved under this name.